

# Constantin Kühne

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✉ Email |  GitHub |  LinkedIn |  Google Scholar

## Education

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### MSc Data Engineering (2023 - present)

*Hasso Plattner Institute/ University of Potsdam*

- Exchange semester at Eindhoven University of Technology (2025)
- Relevant Coursework: Generative AI Models, Deep Learning, Machine Learning for Image Analysis, Reinforcement Learning & Algorithm Discovery, Knowledge Engineering, Advanced Machine Learning Seminars

### BSc Wirtschaftsinformatik/ Business Intelligence (2019 - 2023)

*Hochschule der Bayerischen Wirtschaft*

## Research Experience

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### Semantically Complex Group Object Detection with Vision Language Models (11/2025 - present)

*Hasso Plattner Institute*

- Ongoing master thesis under Prof. Dr. Gerard de Melo at the chair of Artificial Intelligence and Intelligent Systems
- Research on adapting vision language models for semantic group object detection tasks (e.g., detecting a group of people and their actions in street view photos)
- Development of novel neural network architectures and training strategies for this task as well as evaluation on relevant datasets

### Icewafi: A Configurable Data Stream Polluter (11/2024 - 04/2025)

*Hasso Plattner Institute*

- Research assistant role under Dr. Lisa Ehrlinger at the Information Systems chair of Prof. Dr. Felix Naumann
- Research and software development on a framework for data pollution via Apache Flink [\[Link\]](#)
- Evaluation of the effects of data errors on machine learning models

### Beyond Balanced: Learning Static Search Trees with Tree-MDPs (10/2024 - 04/2025)

*Hasso Plattner Institute*

- Research seminar under Alexander Kastius; chair for Artificial Intelligence and Sustainability of Prof. Dr. Ralf Herbrich
- Designed a reinforcement learning framework (Tree-MDP + A2C) to construct static binary search trees for database indexing that adapt to skewed query distributions [\[Link\]](#)
- Achieved up to 40% faster average lookup times on skewed query distributions vs. traditional binary search
- Demonstrated scalability and adaptability to varying data distributions

### TrueSkill/MLPs Beyond Gaussians (10/2024 - 03/2025)

*Hasso Plattner Institute*

- Master's project under Prof. Dr. Ralf Herbrich at the chair for Artificial Intelligence and Sustainability
- Research on more accurate factor graph inference via mixture models (uniform mixtures and Gaussian mixtures) [\[Link\]](#)
- Evaluation of these methods on TrueSkill and Multilayer Perceptrons

### Learning Disentangled Representations with Identifiable Diffusion Models (10/2023 - 09/2024)

*Hasso Plattner Institute*

- Research seminar under Alexander Rakowski and Eshant English at the Digital Health & Machine Learning chair of Prof. Dr. Christoph Lippert
- Research on combining identifiable Variational Autoencoders with Denoising Diffusion Probabilistic Model to gain disentangled latent representations [\[Part1\]](#) [\[Part2\]](#)

### Creation of knowledge graphs based on natural language texts in a domain-specific context (10/2022 - 02/2023)

*Hochschule der Bayerischen Wirtschaft & Infineon Technologies AG*

- Bachelor's thesis on automatically creating knowledge graphs of domain-specific file content
- Fine-tuning of language models (RoBERTa) for triple extraction
- Design and implementation of a semi-automatic triple extraction pipeline via seeds and semi-supervised machine learning

## Industry Experience

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### System Engineer (03/2023 - 08/2023)

*Infineon Technologies AG*

- Implementation and establishment of a Customer Data Platform
- Analysis and value creation out of customer data
- Consolidation and quality assurance of different data sources

### Dual Studies at the Analytics Department (09/2019 - 03/2023)

*Infineon Technologies AG*

- Development, implementation, and operationalization of software solutions in the field of Data Science with a focus on natural language processing
- Creation of semantic search for domain-specific technical documents via Elasticsearch and language models
- Design and implementation of an automatic matching engine for finding and matching funding programmes to fund managers, reducing their workload by up to 50%
- Creation and operationalization of a framework to train machine learning algorithms for the optimization of stock management, resulting in 500K€ (with prospects up to 6.5M€) scrap avoidance and 1600 working hours saved per year

## Technical Skills

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|--------------|---------------------|-----------------|
| • Python     | • PyTorch           | • Docker        |
| • Java       | • Lightning         | • Databricks    |
| • Typescript | • Weights & Biases  | • Elasticsearch |
| • SQL        | • Stable Baselines3 | • Tableau       |

## Awards & Activities

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- Deutschlandstipendium (national merit scholarship), 2023-2025
- ELLIS Summer School on Computer Vision & Machine Learning, Technical University of Munich, 09/2026
  - Selected participant
  - Presented a poster on my master's thesis work

## Publications

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1. Christoph Schininger and Fabian Panse and **Constantin Kühne** and Lisa Ehrlinger. Icewaffl: A Configurable Data Stream Polluter. In Proceedings 28th International Conference on Extending Database Technology (EDBT 2025). <https://doi.org/10.48786/edbt.2025.64>

## Languages

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- German - (C2)
- English - (C1)
- French - (A2)